Lab 4: VibeCoding - Target Shooting

Ankitha Lakshmmi M
PES1UG24CS067

1. Student details

Field	Value
Name	Ankitha Lakshmmi M
SRN	PES1UG24CS067
Repository	https://github.com/ankithalakshmmi/SE-Labs-PES1UG24CS067
Lab folder	Lab-4/ (game in Lab-4/target-shooting/)

2. Deliverables

Deliverable	Where it is	Link / status
a. Before video (10 s)	Lab-4/before.mp4	https://github.com/ankithalakshmmi/SE-Labs-PES1UG24CS067/blob/812f97fad79245309cf4820590b7624fd9ba5a0e/Lab-4/before.mp4
a. After video (10 s)	Lab-4/after.mp4	https://github.com/ankithalakshmmi/SE-Labs-PES1UG24CS067/blob/812f97fad79245309cf4820590b7624fd9ba5a0e/Lab-4/after.mp4
b. Updated code	Lab-4/target-shooting/	https://github.com/ankithalakshmmi/SE-Labs-PES1UG24CS067/tree/main/Lab-4/target-shooting
c. Chat history	Claude share link	https://claude.ai/share/c29ca563-0085-4ccb-8060-0ddf490cee79

3. Commit commands for each task

Run these from the repo root (C:\Users\Ankitha Lakshmmi M\SE-Labs-PES1UG24CS067). Copy the full commit hash from git log --oneline for the last column.

Task	Commands
1. Fix hit detection	`git add Lab-4/target-shooting/game/hit_detection.py` `git commit -m "Fix: hit detection uses circular hitbox"`
2. Moving targets	`git add Lab-4/target-shooting/game/target.py Lab-4/target-shooting/game/game_engine.py` `git commit -m "Feat: moving targets with bounce and two speeds"`
3. Combo scoring	`git add Lab-4/target-shooting/game/game_engine.py` `git commit -m "Feat: combo scoring with multiplier"`
4. Timed round	`git add Lab-4/target-shooting/game/game_engine.py Lab-4/target-shooting/main.py` `git commit -m "Feat: 30s timed round with restart"`

Submission	`git add Lab-4`
	`git commit -m "Lab 4 submission"`
	`git push origin main`

Note: Tasks 2 and 3 both change game_engine.py. If you made one commit for both, use the same hash for both rows and say so here. Check git log --oneline before filling in the hashes.

4. Task status and test results

Task	Status	Test result
1. Fix hit detection	Done	[Write PASS or FAIL after running test_edges.py]
2. Moving targets	Done	Targets bounce off all four walls, stay in the window, and clicks on moving targets register.
3. Combo scoring	Done	Three hits gave scores 1, 3, 6 with combo x2, x3, x4. A miss reset the combo to x1 and kept the score.
4. Timed round	Done (restart by click)	Timer counts down from 30 and reaches 0. Clicks are ignored at 0, "Round over!" and the final score show, and a click after the round ends restarts it.

5. Explanations

Task 1: Fix the hit-detection bug. The original get_bounding_rect built a square from the circle's centre as if it were the top-left corner, so the hitbox was shifted down and to the right. check_hit now measures the distance from the click to the centre with math.hypot and counts a hit only when that distance is within the radius.

Task 2: Moving targets. Each target has horizontal and vertical speeds that are added to its centre every frame. Because x and y are the centre, the wall checks compare the edge of the circle (centre minus or plus the radius) against the window edges. When an edge is passed, the target is pushed back inside and its speed on that axis is reversed. Targets are randomly assigned a slow (1 to 2 px per frame) or fast (4 to 6 px per frame) speed group, so two speeds are visible. Clicks on moving targets still use the circle check from Task 1.

Task 3: Combo scoring. Each hit adds the current multiplier to the score, then raises the multiplier by 1, so consecutive hits score 1, 2, 3, and so on. A miss (clicking empty space) resets the multiplier to 1 but keeps the score. The score and combo are shown on screen during play.

Task 4: 30-second timed round. The round stores the time it started using time.monotonic(), which is real elapsed time, not frame counts. Each frame, the time left is 30 seconds minus the elapsed time, never below zero. When the time left reaches zero, clicks are ignored, the targets stop moving, and a Round over screen shows the final score. A new round resets the score, combo, timer and targets.